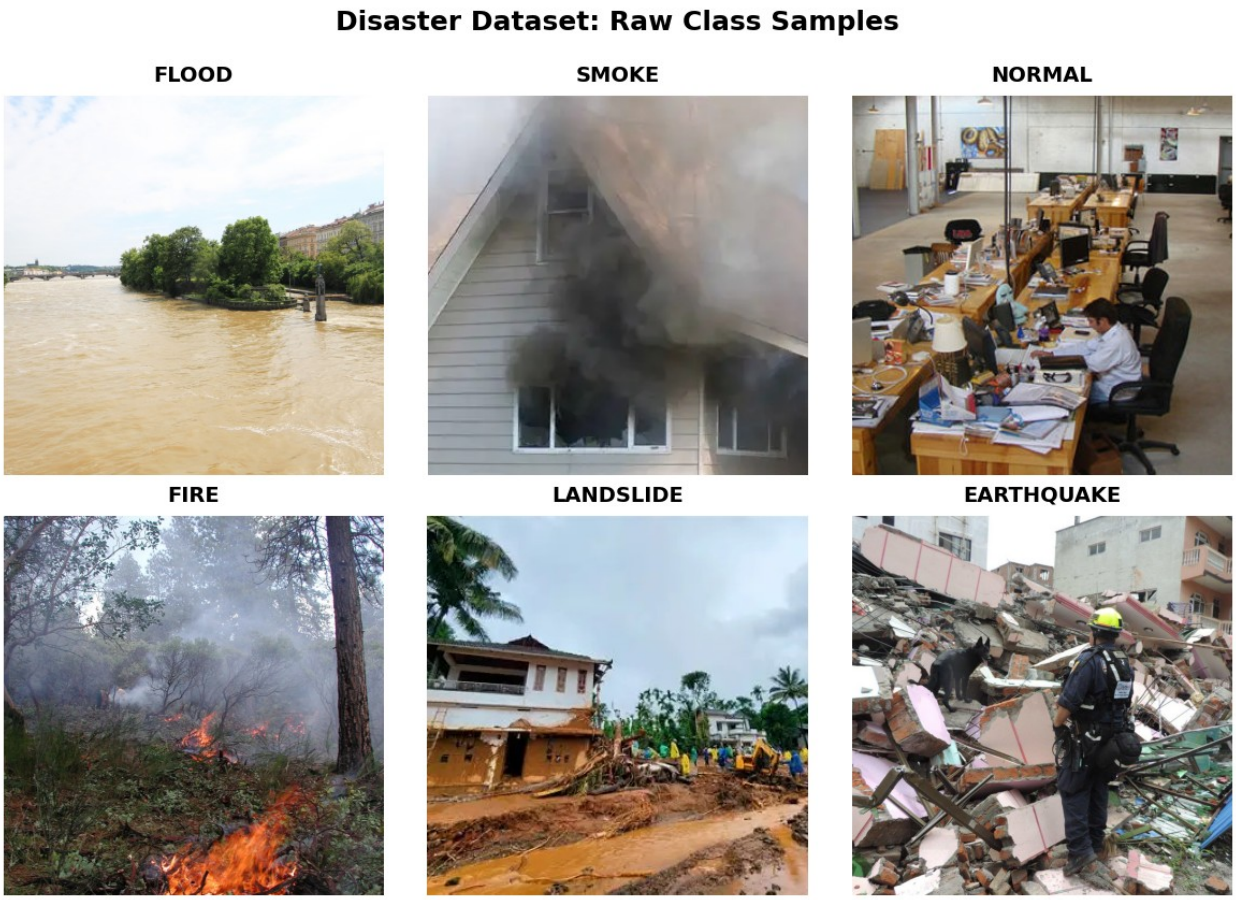
```python
import kagglehub

# Download latest version
path = kagglehub.dataset_download("sarthaktandulje/disaster-damage-5class")

print("Path to dataset files:", path)
```

```
Path to dataset files: /kaggle/input/datasets/sarthaktandulje/disaster-damage-5class
```

```python
# Install required libraries
!pip install -q ultralytics transformers accelerate bitsandbytes gradio kagglehub matplotlib

import kagglehub
import os
import glob
import random
import math
import matplotlib.pyplot as plt
from PIL import Image

print("Downloading dataset...")
# Download the dataset using the path you provided
dataset_path = kagglehub.dataset_download("sarthaktandulje/disaster-damage-5class")
print(f"Dataset downloaded to: {dataset_path}")

# Group images by their class (folder name)
image_paths = glob.glob(os.path.join(dataset_path, "**", "*.jpg"), recursive=True)
image_paths += glob.glob(os.path.join(dataset_path, "**", "*.png"), recursive=True)

dataset_dict = {}
for path in image_paths:
    class_name = os.path.basename(os.path.dirname(path))
    if class_name not in dataset_dict:
        dataset_dict[class_name] = []
    dataset_dict[class_name].append(path)

print(f"\nFound {len(dataset_dict)} classes: {list(dataset_dict.keys())}")

# --- SPLITTING ---
evaluation_set = []
for class_name, paths in dataset_dict.items():
    sampled_paths = random.sample(paths, min(5, len(paths)))
    evaluation_set.extend([(path, class_name) for path in sampled_paths])
```

```
print(f"Created an Evaluation Set of {len(evaluation_set)} images for
pipeline testing.")

# --- UPGRADED VISUALIZATION (EDA) ---
num_classes = len(dataset_dict)
cols = 3
rows = math.ceil(num_classes / cols)

# Create a clean grid
fig, axes = plt.subplots(rows, cols, figsize=(12, 4 * rows))
axes = axes.flatten() # Flatten to easily iterate over the grid

for i, class_name in enumerate(dataset_dict.keys()):
    img_path = random.choice(dataset_dict[class_name])
    img = Image.open(img_path).convert("RGB")

    # Center-crop the image to make it a uniform square
    width, height = img.size
    new_size = min(width, height)
    left = (width - new_size)/2
    top = (height - new_size)/2
    right = (width + new_size)/2
    bottom = (height + new_size)/2
    img_cropped = img.crop((left, top, right, bottom))

    # Plot the cropped image
    axes[i].imshow(img_cropped)
    axes[i].set_title(class_name.upper(), fontsize=14,
fontweight='bold', pad=10)
    axes[i].axis('off')

# Hide any unused subplots (if you had, say, 5 classes in a 6-slot
grid)
for j in range(i + 1, len(axes)):
    axes[j].axis('off')

plt.tight_layout()
plt.suptitle("Disaster Dataset: Raw Class Samples", fontsize=18,
fontweight='bold', y=1.05)
plt.show()
```

```
Downloading dataset...
Dataset downloaded to:
/kaggle/input/datasets/sarthaktandulje/disaster-damage-5class

Found 6 classes: ['flood', 'smoke', 'normal', 'fire', 'landslide',
'earthquake']
Created an Evaluation Set of 30 images for pipeline testing.
```




```
import torch
from ultralytics import YOLO
from transformers import BlipProcessor, BlipForConditionalGeneration
from transformers import AutoTokenizer, AutoModelForCausalLM,
pipeline, BitsAndBytesConfig

print("Loading AI Architectures to GPU...")
device = "cuda" if torch.cuda.is_available() else "cpu"

# 1. YOLOv8
yolo_model = YOLO('yolov8n.pt')

# 2. BLIP
blip_processor = BlipProcessor.from_pretrained("Salesforce/blip-image-
captioning-large")
blip_model =
BlipForConditionalGeneration.from_pretrained("Salesforce/blip-image-
captioning-large").to(device)

# 3. THE UPGRADE: Qwen 2.5 3B (High-Logic LLM)
llm_id = "Qwen/Qwen2.5-3B-Instruct"

# We MUST quantize this larger model to 4-bit so it fits in Kaggle's
```

```
memory alongside YOLO and BLIP
bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_compute_dtype=torch.float16
)

tokenizer = AutoTokenizer.from_pretrained(llm_id)
llm_model = AutoModelForCausalLM.from_pretrained(
    llm_id,
    quantization_config=bnb_config,
    device_map="auto"
)

report_generator = pipeline(
    "text-generation",
    model=llm_model,
    tokenizer=tokenizer,
    max_new_tokens=250,  # <-- INCREASED FOR MORE DESCRIPTIVE OUTPUTS
    temperature=0.2,
    truncation=True
)

print("All models successfully loaded with 4-bit quantization!")
```

```
Loading AI Architectures to GPU...
```

{"model_id":"ab14d36f74f64bb3be4416fb9079f5fb","version_major":2,"version_minor":0}

```
The tied weights mapping and config for this model specifies to tie
text_decoder.bert.embeddings.word_embeddings.weight to
text_decoder.cls.predictions.decoder.weight, but both are present in
the checkpoints, so we will NOT tie them. You should update the config
with `tie_word_embeddings=False` to silence this warning
BlipForConditionalGeneration LOAD REPORT from: Salesforce/blip-image-
captioning-large
Key                                          | Status     |  |
---------------------------------------------+------------+--+-
text_decoder.bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
```

{"model_id":"2e5fc9db6b174c81b522885c976e0d6f","version_major":2,"version_minor":0}

```
All models successfully loaded with 4-bit quantization!
```

```
import cv2
import numpy as np
```

```
import matplotlib.pyplot as plt
from PIL import Image
import random

def generate_disaster_report(img_path, visualize=False):
    # Load and preprocess image
    raw_image = Image.open(img_path).convert("RGB")

    # --- 1. YOLO INFERENCE (Micro-Hazard Detection) ---
    yolo_results = yolo_model(raw_image, verbose=False)

    # Extract annotated image (draws bounding boxes if objects are
found)
    annotated_img_array = yolo_results[0].plot()
    annotated_image =
Image.fromarray(cv2.cvtColor(annotated_img_array, cv2.COLOR_BGR2RGB))

    # Extract object classes
    detected_classes = [yolo_model.names[int(c)] for c in
yolo_results[0].boxes.cls]
    unique_objects = list(set(detected_classes))
    objects_str = ", ".join(unique_objects) if unique_objects else "No
specific objects identified."

    # --- 2. VLM INFERENCE (BLIP Macro-Scene Context) ---
    vlm_inputs = blip_processor(raw_image,
return_tensors="pt").to(device)
    vlm_outputs = blip_model.generate(**vlm_inputs)
    scene_description = blip_processor.decode(vlm_outputs[0],
skip_special_tokens=True)

# --- 3. LLM INFERENCE (Qwen-3B Descriptive Synthesis) ---
    prompt = f"""<|im_start|>system
You are an expert disaster assessment analyst. Write a highly
descriptive, formal damage report based strictly on the provided
visual data. Do not invent objects or hazards that are not mentioned
in the data.<|im_end|>
<|im_start|>user
Visual Data Input:
- YOLO Detected Objects: {objects_str}
- BLIP Scene Context: {scene_description}

Write a detailed assessment report using exactly these three headings:
1. Scene Overview
2. Identified Damage & Hazards
3. Recommended Actions<|im_end|>
<|im_start|>assistant
Assessment Report:
1. Scene Overview:"""
```

```
    # Generate the text
    llm_output = report_generator(prompt)
    generated_text = llm_output[0]['generated_text']

    # Clean up the output using Qwen's specific token markers
    raw_output = generated_text.split("<|im_start|>assistant")[-
1].strip()

    # Fix the prefix to make it look clean in the UI
    if raw_output.startswith("Assessment Report:\n1. Scene
Overview:"):
        final_report = raw_output
    elif raw_output.startswith("1. Scene Overview:"):
        final_report = "Assessment Report:\n" + raw_output
    else:
        final_report = "Assessment Report:\n1. Scene Overview: " +
raw_output

    # --- 4. PIPELINE VISUALIZATION ---
    if visualize:
        plt.figure(figsize=(10, 6))
        plt.imshow(annotated_image)
        plt.axis('off')
        plt.title("YOLO Bounding Box Visualizations", fontsize=14)
        plt.show()

        print(f"--- AI DIAGNOSTICS ---")
        print(f"YOLO Detected: {objects_str}")
        print(f"BLIP Context: {scene_description}")
        print(f"\n--- GENERATED EMERGENCY REPORT ---")
        print(final_report)
        print("-" * 50)

    return annotated_image, objects_str, scene_description,
final_report

# Test the pipeline on a random image from our Evaluation Set to
ensure it works
test_img_path, true_class = random.choice(evaluation_set)
print(f"Testing Pipeline on Class: {true_class}")
_, _, _, _ = generate_disaster_report(test_img_path, visualize=True)
```

```
Testing Pipeline on Class: earthquake
```

```
Both `max_new_tokens` (=250) and `max_length`(=20) seem to have been
set. `max_new_tokens` will take precedence. Please refer to the
documentation for more information.
(https://huggingface.co/docs/transformers/main/en/main_classes/text_ge
neration)
```

YOLO Bounding Box Visualizations

```
--- AI DIAGNOSTICS ---
YOLO Detected: person
BLIP Context: arafed building with a roof that has been torn down

--- GENERATED EMERGENCY REPORT ---
Assessment Report:
1. Scene Overview:
The scene is located at an area near the Arafed building, where a
significant incident involving a person was observed. The immediate
surroundings include the Arafed building, which features a partially
destroyeda roof that has been torn down, exposing the interior
structure.

2. Detailed Damage and Hazards:

 (N/A)
usan't be  to is notvoduced Objectives:
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--------------------------------------------------
```

```python
import gradio as gr

def gradio_interface(uploaded_image):
    temp_path = "gradio_temp.jpg"
    uploaded_image.save(temp_path)

    annotated_image, objects, scene, report =
generate_disaster_report(temp_path, visualize=False)

    markdown_output = f"""
    ### □ Pipeline Diagnostics
    **CNN (YOLO) Output:** {objects}
    **ViT (BLIP) Output:** {scene}

    ### □ Official Damage Report (LLM Output)
    {report}
    """
    return annotated_image, markdown_output

demo = gr.Interface(
    fn=gradio_interface,
    inputs=gr.Image(type="pil", label="Upload Disaster Imagery"),
    outputs=[
        gr.Image(type="pil", label="YOLO Object Detection Analysis"),
        gr.Markdown(label="AI Assessment Output")
    ],
    title="Multi-Modal Disaster Assessment Framework",
    description="Upload a post-disaster image. The system uses
**YOLOv8** (CNN) for micro-hazard detection, **BLIP** (ViT) for macro-
scene understanding, and **TinyLlama** (LLM) for report synthesis.",
    examples=[[path] for path, _ in random.sample(evaluation_set, 3)]
)

demo.launch(share=True)
```

```
* Running on local URL:  http://127.0.0.1:7865
* Running on public URL: https://191bd3a0cfc9f29766.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU
upgrades, run `gradio deploy` from the terminal in the working
directory to deploy to Hugging Face Spaces
(https://huggingface.co/spaces)

<IPython.core.display.HTML object>
```

```
ERROR:    Exception in ASGI application
Traceback (most recent call last):
  File
"/usr/local/lib/python3.12/dist-packages/uvicorn/protocols/http/httpto
ols_impl.py", line 416, in run_asgi
    result = await app(  # type: ignore[func-returns-value]
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/uvicorn/middleware/proxy_head
ers.py", line 60, in __call__
    return await self.app(scope, receive, send)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/fastapi/applications.py",
line 1139, in __call__
    await super().__call__(scope, receive, send)
  File
"/usr/local/lib/python3.12/dist-packages/starlette/applications.py",
line 107, in __call__
    await self.middleware_stack(scope, receive, send)
  File
"/usr/local/lib/python3.12/dist-packages/starlette/middleware/errors.p
y", line 186, in __call__
    raise exc
  File
"/usr/local/lib/python3.12/dist-packages/starlette/middleware/errors.p
y", line 164, in __call__
    await self.app(scope, receive, _send)
  File
"/usr/local/lib/python3.12/dist-packages/gradio/brotli_middleware.py",
line 74, in __call__
    return await self.app(scope, receive, send)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/gradio/route_utils.py", line
882, in __call__
    await self.app(scope, receive, send)
  File
"/usr/local/lib/python3.12/dist-packages/starlette/middleware/exceptio
ns.py", line 63, in __call__
    await wrap_app_handling_exceptions(self.app, conn)(scope, receive,
send)
  File
"/usr/local/lib/python3.12/dist-packages/starlette/_exception_handler.
py", line 53, in wrapped_app
    raise exc
  File
"/usr/local/lib/python3.12/dist-packages/starlette/_exception_handler.
py", line 42, in wrapped_app
    await app(scope, receive, sender)
```

```
  File
"/usr/local/lib/python3.12/dist-packages/fastapi/middleware/asyncexits
tack.py", line 18, in __call__
    await self.app(scope, receive, send)
  File "/usr/local/lib/python3.12/dist-packages/starlette/routing.py",
line 716, in __call__
    await self.middleware_stack(scope, receive, send)
  File "/usr/local/lib/python3.12/dist-packages/starlette/routing.py",
line 736, in app
    await route.handle(scope, receive, send)
  File "/usr/local/lib/python3.12/dist-packages/starlette/routing.py",
line 290, in handle
    await self.app(scope, receive, send)
  File "/usr/local/lib/python3.12/dist-packages/fastapi/routing.py",
line 119, in app
    await wrap_app_handling_exceptions(app, request)(scope, receive,
send)
  File
"/usr/local/lib/python3.12/dist-packages/starlette/_exception_handler.
py", line 53, in wrapped_app
    raise exc
  File
"/usr/local/lib/python3.12/dist-packages/starlette/_exception_handler.
py", line 42, in wrapped_app
    await app(scope, receive, sender)
  File "/usr/local/lib/python3.12/dist-packages/fastapi/routing.py",
line 105, in app
    response = await f(request)
               ^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/fastapi/routing.py",
line 385, in app
    raw_response = await run_endpoint_function(
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/fastapi/routing.py",
line 284, in run_endpoint_function
    return await dependant.call(**values)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/gradio/routes.py",
line 1671, in get_upload_progress
    await asyncio.wait_for(
  File "/usr/lib/python3.12/asyncio/tasks.py", line 520, in wait_for
    return await fut
           ^^^^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/gradio/route_utils.py", line
528, in is_tracked
    return await self._signals[upload_id].wait()
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/asyncio/locks.py", line 209, in wait
    fut = self._get_loop().create_future()
```

```
          ^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/asyncio/mixins.py", line 20, in _get_loop
    raise RuntimeError(f'{self!r} is bound to a different event loop')
RuntimeError: <asyncio.locks.Event object at 0x7a24475426f0 [unset]>
is bound to a different event loop
Both `max_new_tokens` (=250) and `max_length`(=20) seem to have been
set. `max_new_tokens` will take precedence. Please refer to the
documentation for more information.
(https://huggingface.co/docs/transformers/main/en/main_classes/text_ge
neration)
```